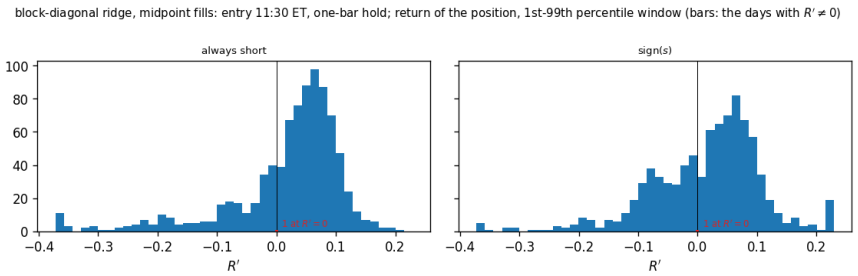
0DTE nearest-OTM straddle: per-rule return summary, entry 11:30 ET, one-bar hold, models compared on the same 866 days

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	865	0.016	0.114	-1.06	0.21	-3.37	19.30	4.05	2.19	0	0.0	-2.13
sign(s) (± 1 *on the sign of* s)												
baseline (HAR + calendar OLS)	865	0.019	0.114	-0.90	1.06	-0.16	17.81	4.80	2.59	231	26.7	-1.82
block-diagonal ridge	865	0.017	0.114	-0.65	1.06	1.13	16.89	4.49	2.42	252	29.1	-2.01
block-diagonal ridge, without the FOMC columns	865	0.017	0.114	-0.65	1.06	1.14	16.86	4.38	2.36	265	30.6	-2.07
LightGBM†	865	0.020	0.114	-0.65	1.06	1.06	17.08	5.11	2.76	221	25.5	-1.68
XGBoost†	865	0.020	0.114	-0.65	1.06	1.03	17.11	5.13	2.77	213	24.6	-1.67
lasso (causally tuned)†	865	0.018	0.114	-0.65	1.06	1.05	16.94	4.51	2.43	238	27.5	-1.99
lasso ($\alpha = 10^{-4}$)	865	0.016	0.114	-0.65	1.06	1.16	16.81	4.17	2.25	271	31.3	-2.17
elastic net (causally tuned)†	865	0.018	0.114	-0.65	1.06	1.05	16.95	4.54	2.45	236	27.3	-1.98



Distribution of the return of the position taken at 11:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 11:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 865 expiration days (the other 1 of the deck's 866 days have no quoted straddle at this stamp), 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged. **Straddle** here means the nearest out-of-the-money call plus the nearest out-of-the-money put, same-day expiry, bought or sold as one position. One expiration day = one return; mid fill. Every t here is the plain $t = \sqrt{n} \cdot \text{mean/std}$, with no autocorrelation correction. "ex. kurt" is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. **Two Sharpes.** Both multiply by $\sqrt{252}$ because an expiration day is the unit of time (same year-length as the paper). They are not the same object. On a *single-clock* page, $\text{Sharpe}_{\text{clk}} = (\text{mean/std}) \times \sqrt{252}$ of that clock's series: one 30-minute hold per expiration day. It answers "if I only entered at this clock, what is my annual Sharpe?" It is *not* the book's $\text{Sharpe}_{\text{ann}}$. On the *pooled* page, $\text{Sharpe}_{\text{ann}} = (\text{mean/std}) \times \sqrt{252}$ of $R_d^{day} = \sum_t R'_{d,t}$, the sum of the day's twelve holds. That reserved name is the book's daily P&L after pooling the session. At 15:30 the two coincide (one trade that day); they do not on 10:00–15:00 or on the pooled page. $\text{Sharpe}^{\times}$ is the same convention at the crossed spread. Every other column is daily. n_{buy} is the number of expiration days with position > 0 (always-short is 0; on the pooled page it is days the sum of the twelve positions is positive); %buy is $100 \cdot n_{\text{buy}}/n$, the notebook's column. The other column this table adds to the deck's is $\text{Sharpe}^{\times}_{\text{ann}}$, the same rule's annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck's.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned). † marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is $s_t = \widehat{RV}_t - \text{IV}^2_{\text{hr}} h_t w_t$: the forecast against the implied variance of the same window. w_t is the expanding mean of each prior day's remaining-session share of realized variance, $\text{RV}_t / \sum_{s \geq t} \text{RV}_s$ (in-fit panel history back to 2001), not the ratio of trailing clock-means. On the bars whose vendor implied volatility is a censored solver node the deck's treatment is used here too — the volatility that reproduces the straddle midpoint, recovered by bisection over the remaining h_t hours — so every bar carries a signal and no bar sits flat.